Final comparison: legacy vs hybrid PQC for every approach

StarFive VisionFive 2 (4 × SiFive U74 RISC-V, Debian, OpenSSL 3.5.9). All values are milliseconds per connection, **1000 connections per mode**; lower is better. Legacy = X25519 + RSA-3072 (the baseline connection). Hybrid = X25519 + ML-KEM-768 key exchange, plus ECDSA P-256 + ML-DSA-44 authentication on our own server. Detailed reports, raw data and scripts: `results/`.

1. Our own server (full hybrid: key exchange + authentication, local network)

Setting	Legacy best	Legacy median	Legacy worst	Hybrid best	Hybrid median	Hybrid worst
Default CPU (baseline)	14.8	**21.6**	31.3	20.5	**28.0**	40.1
CPU bump, 750 MHz minimum	8.9	**15.7**	21.7	13.9	**16.2**	24.3
Bit manipulation (Zba/Zbb)	11.7	**21.1**	33.3	19.2	**26.6**	38.4
Multicore (2nd core for ML-DSA)	15.4	**21.7**	35.3	17.5	**24.6**	38.1
All three combined	9.0	**15.4**	21.4	**12.2**	**14.5**	**26.6**

2. FOCAL's server (hybrid key exchange, over the internet)

Setting	Legacy best	Legacy median	Legacy worst	Hybrid best	Hybrid median	Hybrid worst
Default CPU (baseline)	31.6	**39.9**	524.5	36.3	**45.1**	359.3
CPU bump, 750 MHz minimum	26.1	**37.2**	303.6	29.3	**40.3**	200.4
Bit manipulation (Zba/Zbb)	31.0	**41.3**	274.6	34.0	**47.5**	547.6
Multicore	not possible: FOCAL's server supports the hybrid key exchange, but only RSA authentication					
Combined (clock + bitmanip; multicore n/a)	25.9	**33.0**	355.0	**27.7**	**35.6**	**140.8**

On FOCAL's server, bit manipulation shows no clear change: the internet delay (about 30 ms) hides a 1–2 ms gain.

3. Final comparison (baseline and combined measured in the same session: the fairest)

	Best	Median	Worst
Own server, legacy baseline	12.6	**21.0**	35.1
Own server, **hybrid combined**	**12.2**	**14.5**	**26.6**
FOCAL, legacy baseline	28.2	**38.4**	354.3
FOCAL, **hybrid combined** (clock + bitmanip)	**27.7**	**35.6**	**140.8**

"Combined" = all three approaches on our own server. On FOCAL's server only the CPU clock and bit manipulation apply, because it uses RSA authentication, so there's no ML-DSA signature to verify on a second

core.

Hybrid combined beats the legacy baseline on best, median and worst: –31 % median on our own server (faster in 20/20 test blocks) and –7 % median on FOCAL's server (17/20 blocks). 8,000 connections, 0 failures, +1.3 °C idle temperature.

4. How to read best / median / worst

- **Best:** the fastest single connection, what the board can do with no disturbance.
- **Median:** the typical connection. This is the headline number and the fairest one.
- **Worst:** a single connection out of 1000, so it's random.
- On **our own server** it's meaningful (local network, little noise): hybrid worst 40.1 → 26.6 ms, so the slow cases improved too.
- On **FOCAL's server**, worst values (140–548 ms) come from one-off internet delays, not from the crypto. Don't compare them as results; that's why we report the median.

5. What each approach does to the hybrid median (own server)

Approach	Hybrid median	Change	Why
Default CPU (baseline)	28.0		CPU idles at 312.5 MHz; a 20 ms handshake ends before it speeds up
CPU bump, 750 MHz minimum	16.2	**–11.6 ms**	Biggest win: the crypto no longer starts at the lowest clock
Bit manipulation	26.6	–1.6 ms	Single-instruction rotations: SHA-3 +50 %, ML-KEM up to –19 %
Multicore	24.6	–3.6 ms	ECDSA and ML-DSA checked at the same time on two cores
All three combined	**14.5**	**–13.6 ms**	**Below legacy's 21: 31 % faster than the baseline**

6. Quick memory

- **Own server, hybrid median:** 28 → 16.2 (bump) · 26.6 (bitmanip) · 24.6 (multicore) · **14.5 (all)**, vs legacy **21**.
- **Own server, hybrid worst:** 40 → **26.6**.
- **FOCAL, hybrid median:** 45 → **35.6**, vs legacy **38–40**.

Notes

- Each single-approach row (1 and 2) comes from its own test session, so its legacy values differ slightly (21.1–21.7 ms). Section 3 is the cleanest comparison, because baseline and combined were alternated block by block in one session.
- Multicore only changes hybrid: legacy has a single signature to check, so its row is just that session's reference.
- Sources (all in `results/`): `01_baseline_default_clock/baseline_stats_1000.txt`, `02_dynamic_cpu_freq/dynamic_cpu_freq_confirm_stats.txt` (D1), `03_bitmanip_zba_zbb/bitmanip_e2e_stats.txt`, `04_multicore/multicore_e2e_stats.txt` (v2_par), `05_combined_all_approaches/combined_stats.txt`.

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